In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [4]:
```
data = pd.read_csv("C:/Users/Ganesh/Downloads/Social_Network_Ads.csv")
```

In [5]:
```
data
```

Out[5]:

| | User ID | Gender | Age | EstimatedSalary | Purchased |
|---|---|---|---|---|---|
| **0** | 15624510 | Male | 19 | 19000 | 0 |
| **1** | 15810944 | Male | 35 | 20000 | 0 |
| **2** | 15668575 | Female | 26 | 43000 | 0 |
| **3** | 15603246 | Female | 27 | 57000 | 0 |
| **4** | 15804002 | Male | 19 | 76000 | 0 |
| **...** | ... | ... | ... | ... | ... |
| **395** | 15691863 | Female | 46 | 41000 | 1 |
| **396** | 15706071 | Male | 51 | 23000 | 1 |
| **397** | 15654296 | Female | 50 | 20000 | 1 |
| **398** | 15755018 | Male | 36 | 33000 | 0 |
| **399** | 15594041 | Female | 49 | 36000 | 1 |

400 rows × 5 columns

In [6]:
```
data.head(5)
```

Out[6]:

| | User ID | Gender | Age | EstimatedSalary | Purchased |
|---|---|---|---|---|---|
| **0** | 15624510 | Male | 19 | 19000 | 0 |
| **1** | 15810944 | Male | 35 | 20000 | 0 |
| **2** | 15668575 | Female | 26 | 43000 | 0 |
| **3** | 15603246 | Female | 27 | 57000 | 0 |
| **4** | 15804002 | Male | 19 | 76000 | 0 |

In [7]:
```
data.tail()
```

Out[7]:

| | User ID | Gender | Age | EstimatedSalary | Purchased |
|---|---|---|---|---|---|
| **395** | 15691863 | Female | 46 | 41000 | 1 |
| **396** | 15706071 | Male | 51 | 23000 | 1 |
| **397** | 15654296 | Female | 50 | 20000 | 1 |
| **398** | 15755018 | Male | 36 | 33000 | 0 |
| **399** | 15594041 | Female | 49 | 36000 | 1 |

In [8]:
```
data.shape
```

Out[8]:
```
(400, 5)
```

In [9]:
```
data.columns
```

Out[9]:
```
Index(['User ID', 'Gender', 'Age', 'EstimatedSalary', 'Purchased'], dtype='objec
t')
```

In [10]:
```
data.describe()
```

Out[10]:

| | User ID | Age | EstimatedSalary | Purchased |
|---|---|---|---|---|
| **count** | 4.000000e+02 | 400.000000 | 400.000000 | 400.000000 |
| **mean** | 1.569154e+07 | 37.655000 | 69742.500000 | 0.357500 |
| **std** | 7.165832e+04 | 10.482877 | 34096.960282 | 0.479864 |
| **min** | 1.556669e+07 | 18.000000 | 15000.000000 | 0.000000 |
| **25%** | 1.562676e+07 | 29.750000 | 43000.000000 | 0.000000 |
| **50%** | 1.569434e+07 | 37.000000 | 70000.000000 | 0.000000 |
| **75%** | 1.575036e+07 | 46.000000 | 88000.000000 | 1.000000 |
| **max** | 1.581524e+07 | 60.000000 | 150000.000000 | 1.000000 |

In [11]:
```
data.isnull().sum()
```

Out[11]:
```
User ID            0
Gender             0
Age                0
EstimatedSalary    0
Purchased          0
dtype: int64
```

In [12]:
```
data.iloc[:,2:4]
```

Out[12]:

| | Age | EstimatedSalary |
|---|---|---|
| **0** | 19 | 19000 |
| **1** | 35 | 20000 |
| **2** | 26 | 43000 |
| **3** | 27 | 57000 |
| **4** | 19 | 76000 |
| **...** | ... | ... |
| **395** | 46 | 41000 |
| **396** | 51 | 23000 |
| **397** | 50 | 20000 |
| **398** | 36 | 33000 |
| **399** | 49 | 36000 |

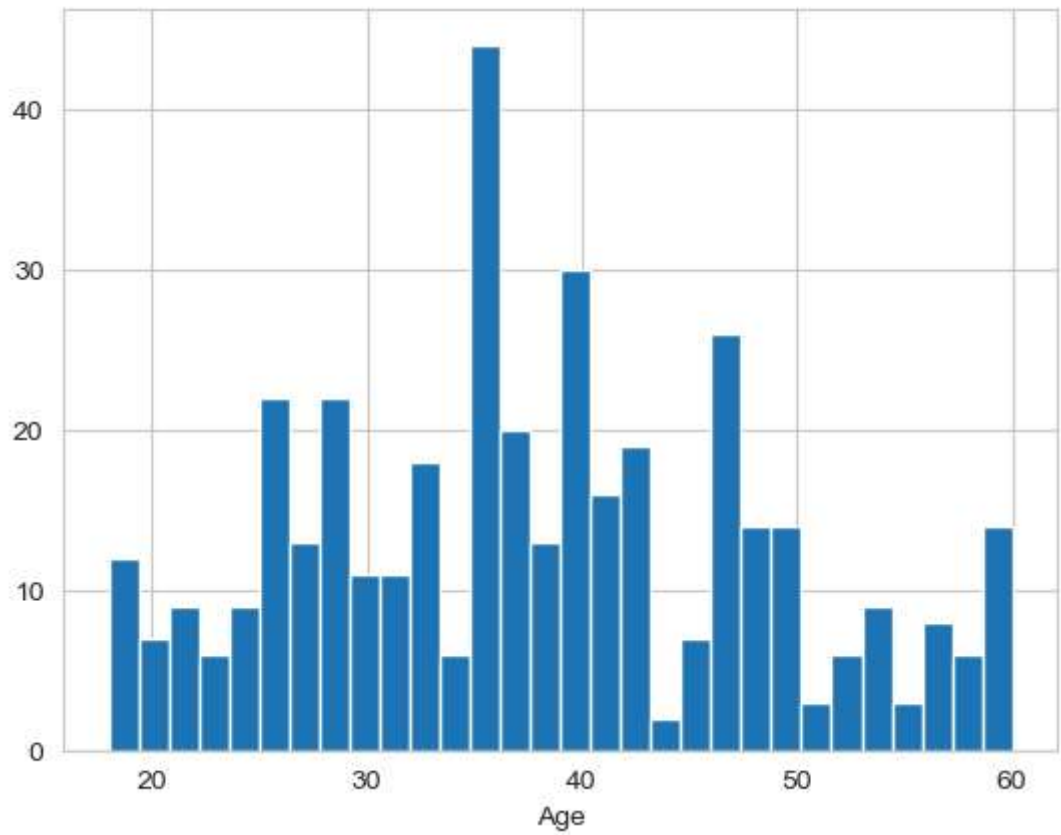
400 rows × 2 columns

In [13]:
```
ndata = data.iloc[:,2:4].values
```

In [14]:
```
import seaborn as sns
```

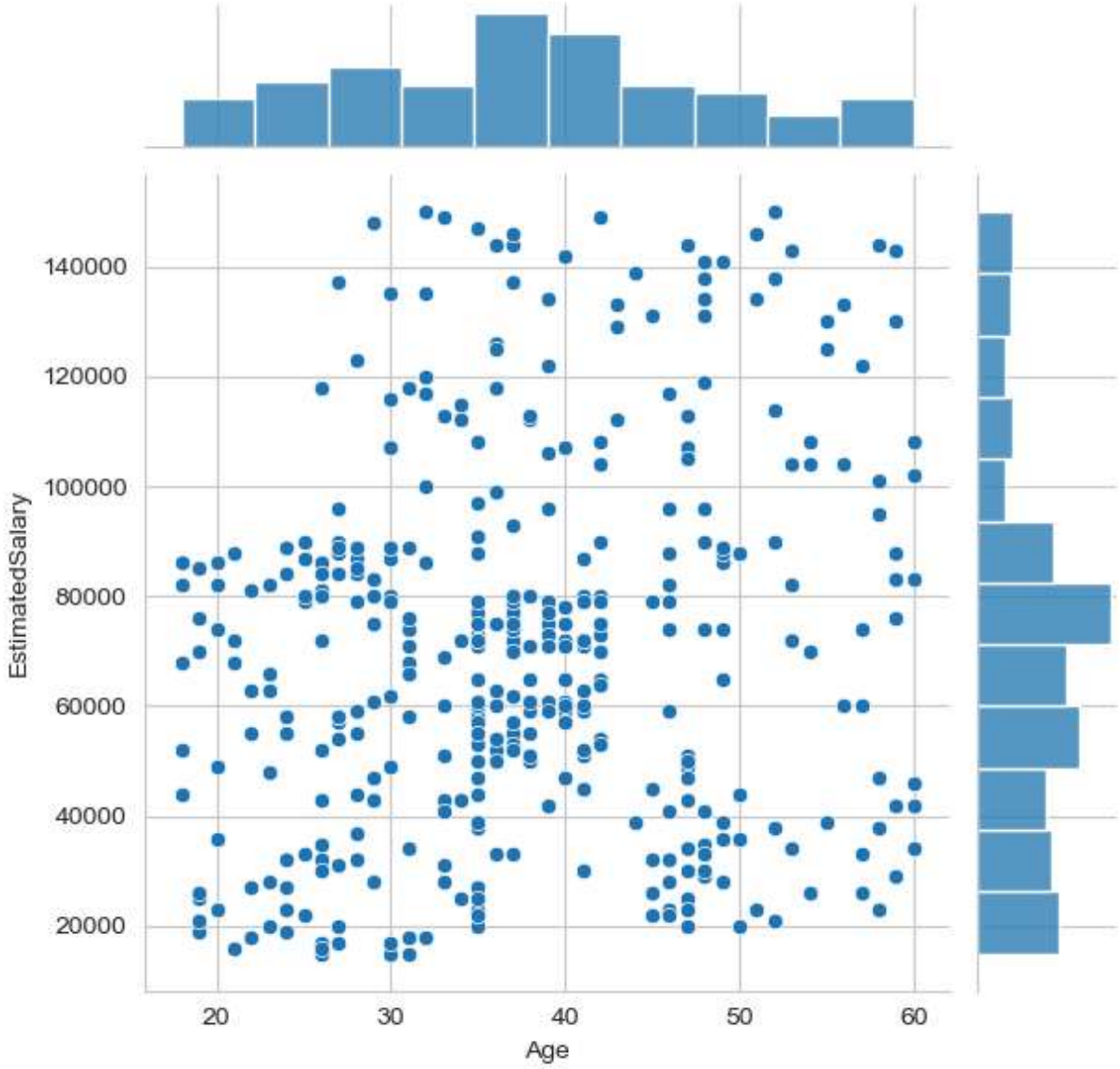
In [15]:
```
sns.set_style('whitegrid')
data['Age'].hist(bins=30)
plt.xlabel('Age')
```

Out[15]:
```
Text(0.5, 0, 'Age')
```



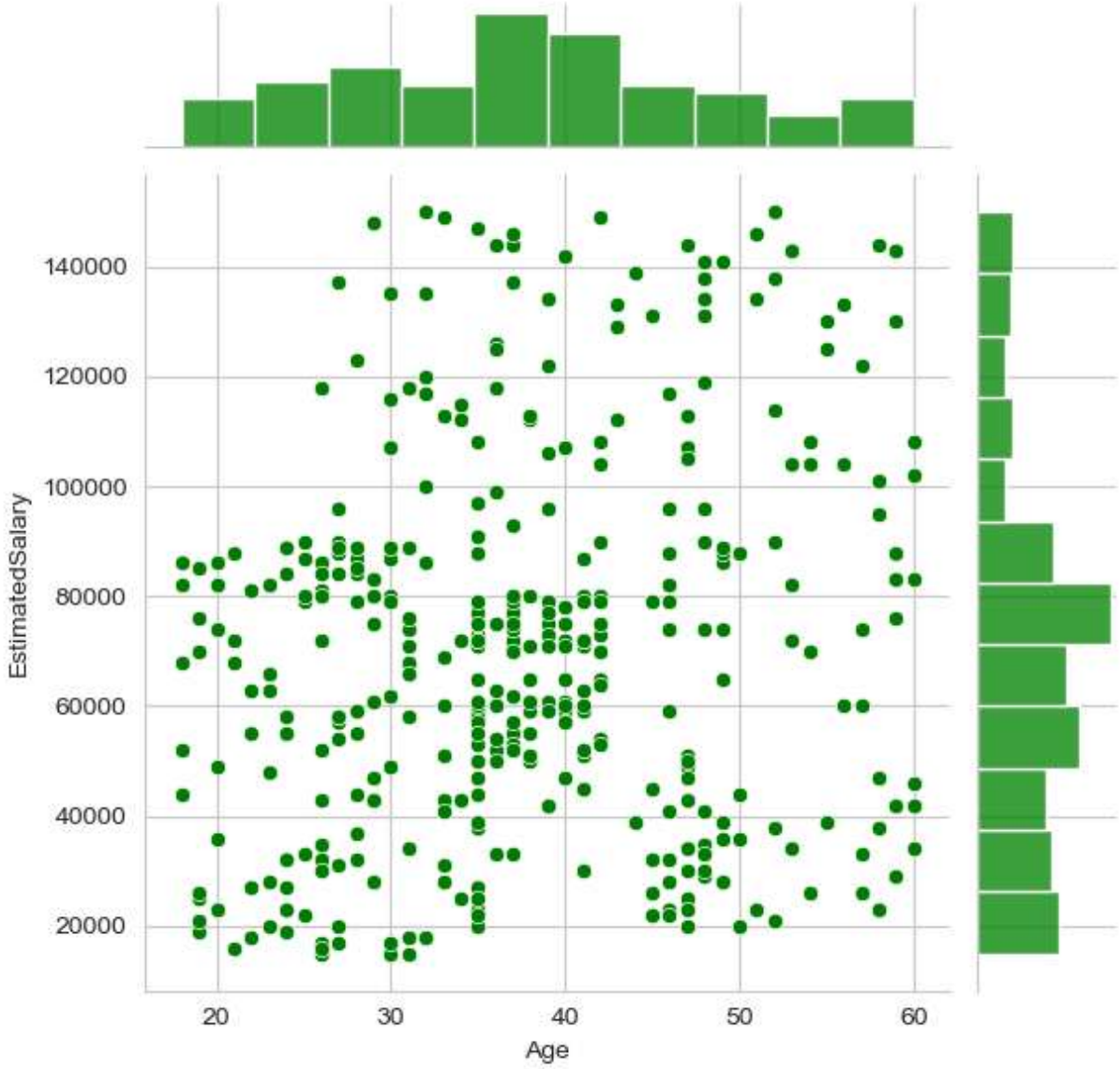

In [16]:
```
sns.jointplot(x='Age', y='EstimatedSalary', data = data)
```

Out[16]:
```
<seaborn.axisgrid.JointGrid at 0x1fe4ef1cad0>
```



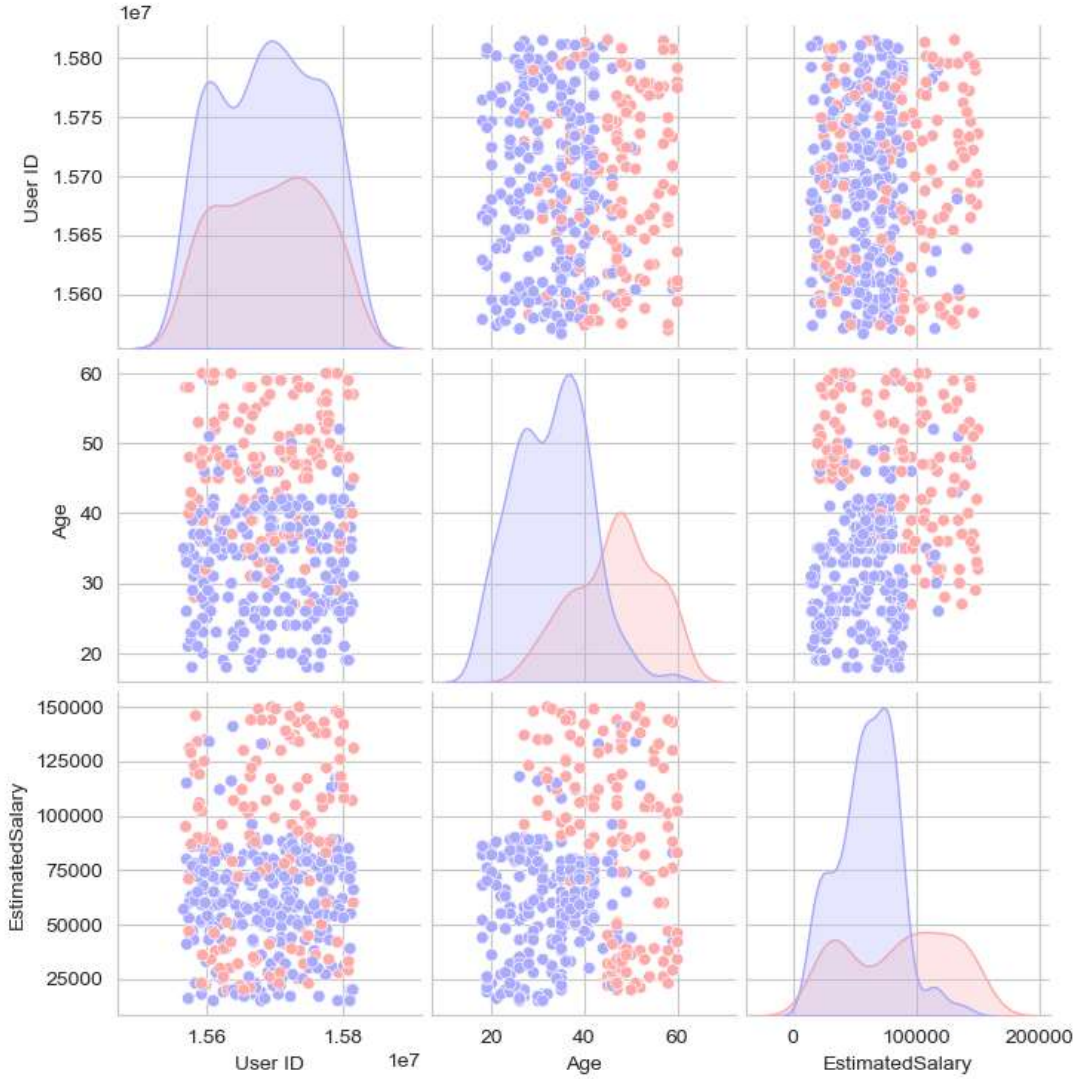

In [17]:
```
sns.jointplot(x='Age',y='EstimatedSalary',data= data,color='green')
```

Out[17]:
```
<seaborn.axisgrid.JointGrid at 0x1fe4f1ebb10>
```




In [18]:
```
sns.pairplot(data,hue='Purchased',palette='bwr')
```

Out[18]:
```
<seaborn.axisgrid.PairGrid at 0x1fe4ef1d160>
```




In [19]:
```
from sklearn.model_selection import train_test_split
```

In [20]:
```
X = data[['Age','EstimatedSalary']]
y = data['Purchased']
```

In [21]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_st
```

In [22]:
```
X_train.shape
```

Out[22]: (300, 2)

In [23]:
```
X_test.shape
```

Out[23]: (100, 2)

In [24]:
```
y_train.shape
```

Out[24]: (300,)

In [25]:
```
y_test.shape
```

Out[25]: (100,)

In [26]:
```
from sklearn.linear_model import LogisticRegression
```

In [27]:
```
logmodel = LogisticRegression()
logmodel.fit(X_train,y_train)
```

Out[27]:
```
▾ LogisticRegression
▶ Parameters
```

In [31]:
```
predictions = logmodel.predict(X_test)
```

In [32]:
```
from sklearn.metrics import classification_report
```

In [33]:
```
print(classification_report(y_test,predictions))
```

```
              precision    recall  f1-score   support

           0       0.89      0.96      0.92        68
           1       0.89      0.75      0.81        32

    accuracy                           0.89       100
   macro avg       0.89      0.85      0.87       100
weighted avg       0.89      0.89      0.89       100
```

In [34]:
```
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, predictions)
print(cm)
```

```
[[65  3]
 [ 8 24]]
```